

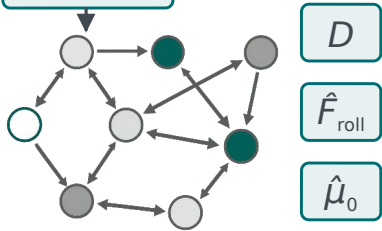
FLOW SPACE

budgets b_k

$b_k, k = 1 \dots K$

2 Flow graph

discretise



$\omega^{(k)}, \bar{c}_k$

3 Capacity compile

$S^{(1)}$

$S^{(2)}$

$\vdots$

$S^{(K)}$

$$S^{(k)} = b_k \omega^{(k)} / \bar{c}_k$$

$S^{(1 \dots K)}$

4 Compose

hard min
(certificate)

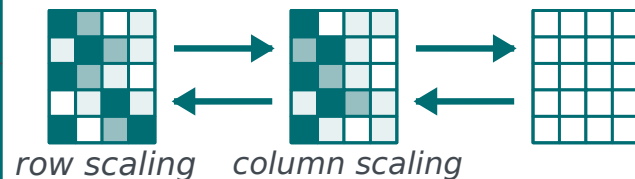
softmin T_c

S
(capacity ceiling)

5 γ -Flow Sinkhorn

$$\min \langle D, F \rangle - \varepsilon H(F)$$

ε



$$0 \leq F \leq S \quad F \mathbf{1} - \gamma F^T \mathbf{1} = (1 - \gamma) \mu_0$$

F^*

transitions, rewards,
 K costs

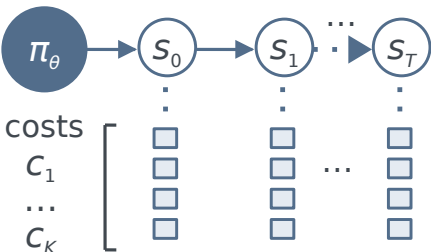
$D, \hat{\mu}_0$

gates

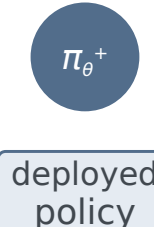
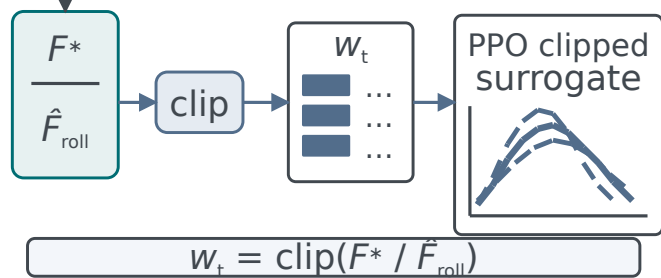
F^*

POLICY SPACE

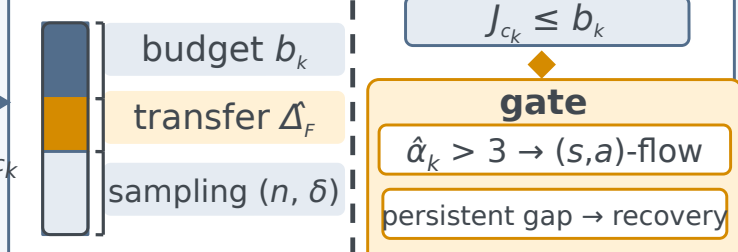
1 Policy rollout



6 Flow-weighted PPO



7 Deployed-policy audit



next rollout